FutureScout: Fourth-Down Decision Value and Quarterback Salary Efficiency in the NFL

Project Proposal · Data Science Capstone · Fall 2026

1. The Team

Pranav Kaliaperumal (pranavkumar.kaliaperumal@ucdenver.edu) · solo project. I will own every stage of the work: data acquisition, cleaning, modeling, visualization, the interactive site, and the final write-up. I already have a working Python pipeline for team-level NFL data from coursework, so I can spend most of my time on the new play-by-play modeling. Planned split of effort: roughly 40% modeling, 30% data engineering, 20% visualization and site, 10% writing.

2. Problem Statement and Background

NFL coaches face two recurring resource-allocation problems. The first happens dozens of times every Sunday: on fourth down, should the offense go for it, punt, or attempt a field goal? The second happens every offseason: how much of the salary cap should a team commit to its quarterback? FutureScout attacks both questions with the same machinery, probabilistic models trained on historical outcomes, and asks whether actual team behavior matches what the numbers recommend.

The fourth-down question has a long history. Carter and Machol used expected points to value field position as early as 1971[1]. Romer showed that teams kick far more often than an expected-value calculation justifies, and his paper became a reference point for the modern analytics movement[2]. Yam and Lopez later estimated that conservative fourth-down choices cost teams a measurable fraction of wins per season[3]. Coaches have moved toward aggression in recent years, but league-wide behavior still varies wildly by team, score, and field position. I want to quantify that gap with a win-probability model rather than expected points, because win probability folds score and clock into the decision directly.

The salary question has a shorter academic history but a louder public one. Most published work stops at correlating spending with production. I want to push past correlation into a value framework: which teams squeeze the most quarterback performance out of each cap dollar, and how much does quarterback spending actually matter once efficiency enters the model? Front offices care about this every March; the rookie wage scale made cheap quarterback labor the most discussed market inefficiency in the sport.

3. Data Sources

I will use two data sources, both already in hand.

The first is the nflverse play-by-play repository (the data backbone of the nflfastR project)[4]. It publishes one CSV per season covering every NFL play since 1999 with more than 300 columns: down, distance, field position, score, timeouts, play type, result, and pre-computed expected points. I pulled the 2015

through 2024 regular seasons, about 460,000 plays across 2,623 games. Ten seasons keep the sample large while limiting drift from rule changes and the league's shifting pass-run balance.

The second source is a team-season dataset I assembled from Pro Football Reference passing and rushing tables plus team salary figures by position group. It covers 320 team-seasons (2015-2025, excluding 2021, for which a complete salary table was unavailable) with team-weighted QB QBR, a composite running back rating built from success rate, yards per attempt, and scoring, position-group salary spending, wins, and playoff indicators. QBR values are weighted across each team's quarterbacks by pass attempts; the running back rating is weighted by rushing attempts.

Cleaning work up front: restrict the play-by-play to regular-season plays with a defined possession team; drop two-point conversions, kneels, and aborted plays from the modeling frames; rebuild score differential and clock variables from the possession team's perspective; and construct post-play states (field position after a turnover on downs, after a made or missed field goal, after a punt) for the decision engine. For the salary data I will normalize spending within each season, since the cap rose by more than 60% over the sample window and raw dollars are not comparable across years.

4. Goals of the Analysis

For the fourth-down module I plan to test three hypotheses:

- **H1.** On fourth-and-short (2 yards or fewer) between midfield and the opponent's 35, going for it yields higher expected win probability than punting for the majority of game states.
- **H2.** Teams go for it less often than the model recommends, and the gap is largest in the region Romer identified, short yardage near midfield[2].
- **H3.** The model's recommendation flips toward aggression late in games when trailing, and toward conservatism when leading, by an amount large enough to reorder team aggressiveness rankings across quarters.

For the quarterback value module I define performance above expected as the residual of QBR regressed on within-season salary share, and value as that residual per dollar committed:

$$\text{QB Value}_t = \frac{\text{QBR}_t - \widehat{\text{QBR}}(\text{salary share}_t)}{\text{QB salary share}_t}$$

I will then model playoff qualification as (1)

$$P(\text{Playoffs}) = f(\text{QB QBR},\ \text{QB salary share},\ \text{RB rating},\ \text{RB salary share},\ \text{seaso}$$

and test two more hypotheses: (2)

- **H4.** QB efficiency (QBR) predicts playoff qualification far better than QB spending; adding spending on top of QBR improves out-of-sample log loss by less than 5%.
- **H5.** Playoff teams cluster below the league median in QB salary share, consistent with the rookie-contract advantage.

Success metrics: the win-probability model should beat a logistic baseline on held-out log loss and Brier score and stay calibrated within 3 percentage points across deciles; the playoff classifier should reach cross-validated AUC above 0.80, which matches published team-success models.

5. Tools

The pipeline runs in Python. I use pandas and NumPy for cleaning and feature engineering, scikit-learn for the models (logistic regression baselines, histogram gradient boosting for the main win-probability and conversion models, plus random forest and regularized logistic regression for the playoff comparison), and Matplotlib for figures. Model evaluation uses a time-based split: train on 2015-2022, test on 2023-2024, so leakage between seasons stays impossible. A single setup script downloads the raw play-by-play files from the nflverse release server and reproduces every processed dataset and figure. The final site is static HTML/CSS/JavaScript, with a client-side decision calculator driven by exported logistic model coefficients.

6. Data Products

- A fourth-down decision heatmap over field position and yards to go, for a neutral game state and for late-game states.
- Win-probability curves around fourth-down plays and calibration plots for every probabilistic model.
- Feature-importance rankings for the win-probability and conversion models.
- Team aggressiveness rankings for 2015-2024: actual go rate versus model-recommended go rate, and estimated win probability left on the table per season.
- An interactive calculator that takes score, clock, field position, and distance and returns GO / FG / PUNT with expected win probability for each option.
- Quarterback value rankings by team-season, an efficiency-frontier scatter of salary share against QBR, and a model-comparison table for the playoff classifier.

References

- Carter, V., & Machol, R. E. (1971). Operations research on football. *Operations Research, 19*(2), 541-544.
- Romer, D. (2006). Do firms maximize? Evidence from professional football. *Journal of Political Economy, 114*(2), 340-365.
- Yam, D. R., & Lopez, M. J. (2019). Quantifying the causal effects of conservative fourth down decision making in the National Football League. *Journal of Quantitative Analysis in Sports, 15*(3), 171-186.
- Carl, S., Baldwin, B., Ho, T., Sharpe, L., & Edwards, J. (2020). *nflfastR: An R package for NFL play-by-play data* [Computer software]. nflverse. https://www.nflfastr.com